

Supply & Demand — Must-Know

The essentials you cannot walk into the exam without

- › Solve for equilibrium · surplus · deadweight loss
- › Elasticity & the revenue rule
- › Taxes: the wedge & who really pays
- › Competitive firm ($P = MC$) vs market power (markup)

The 12 things you must know

1. **Demand slopes down** (= willingness to pay), **supply slopes up** (= marginal cost / willingness to accept). They cross at the **equilibrium** (P^* , Q^*).
2. **To find equilibrium: set quantity demanded = quantity supplied, solve for p, then plug back in for q.**
3. **Consumer surplus** = triangle **under demand, above price**. **Producer surplus** = triangle **above supply, below price**. Both are $\frac{1}{2} \cdot \text{base} \cdot \text{height}$.
4. **Elasticity** $\epsilon = (\% \Delta Q) / (\% \Delta P)$ in absolute value. $\epsilon > 1$ elastic, $\epsilon < 1$ inelastic. **The slope coefficient is NOT the elasticity.**
5. **Inelastic demand:** raising price **raises** revenue. **Elastic demand:** raising price **lowers** revenue.
6. **A shift's split into price vs quantity depends on elasticity.** More elastic demand $\rightarrow$ shift moves quantity a lot, price a little.
7. **A tax drives a wedge:** buyer price rises, seller price falls, quantity falls. **It doesn't matter who legally pays** — same outcome.
8. **Tax incidence: the more INELASTIC side bears the bigger share.** A tax creates a **deadweight loss**.
9. **Competitive (price-taking) firm: produce where $P = MC$.** Its supply curve is its MC curve (above AC). Exit if $P < AC$.
10. **Long run under competition: entry drives price to $P = MC = AC$** (zero economic profit).
11. **Firm with market power: $MR < P$.** For linear demand, **MR has the same intercept but is twice as steep**. Produce where **$MR = MC$** , then read the price up on the demand curve (so **$P > MC$**).
12. **Markup rule: $(P - MC)/P = 1/\epsilon$.** More elastic demand $\rightarrow$ smaller markup.
Market power $\rightarrow$ deadweight loss; competition removes it.

CORE SKILL 1

Solve for the Equilibrium

THE METHOD (WORKS EVERY TIME)

Step 1. Write demand and supply as quantities: $q_D(p)$ and $q_S(p)$.

Step 2. Set $q_D = q_S$ and solve for the price p^* .

Step 3. Put p^* back into either equation to get q^* .

WORKED EXAMPLE

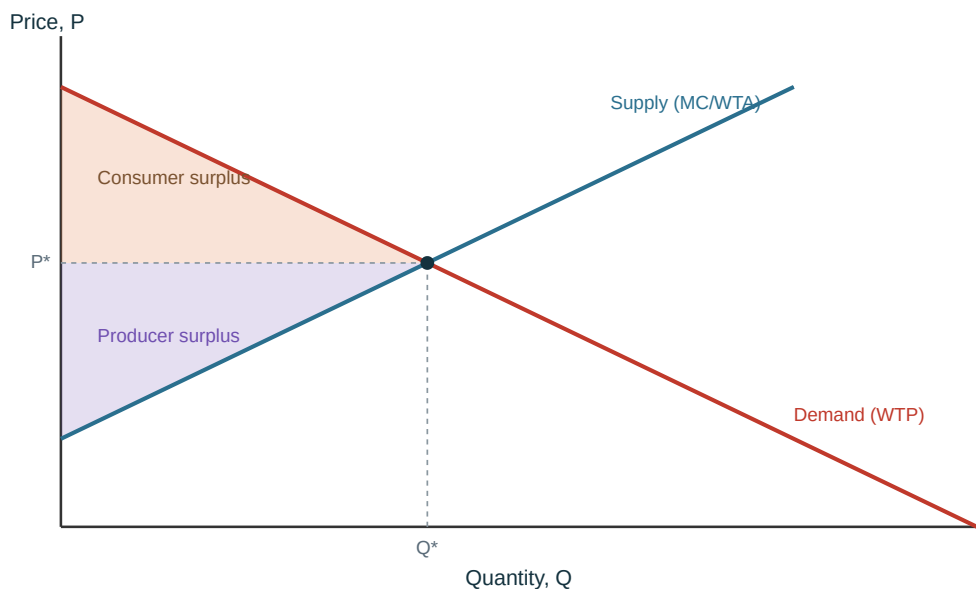
Demand $q = 200 - 5p$, supply $q = -50 + 20p$.

Set equal: $200 - 5p = -50 + 20p \rightarrow 250 = 25p \rightarrow p^* = 10$. Then $q^* = 200 - 5(10) = 150$.

EXAM TRAP

"**Normal slopes**" means demand down, supply up. If a question says supply is **perfectly inelastic** it's a **vertical** line (fixed quantity); **perfectly elastic** is a **horizontal** line (fixed price). Excess supply (price too high) pushes price **down**; excess demand (price too low) pushes it **up**.

Consumer & producer surplus — the master picture



The one diagram to memorise. Consumer surplus = area **under demand, above P^*** . Producer surplus = area **above supply, below P^*** . Total surplus (CS+PS) is **maximised** at the competitive equilibrium — no deadweight loss.

SURPLUS FORMULAS (STRAIGHT-LINE CURVES)

CS = $\frac{1}{2} \times Q^* \times (\text{demand choke price} - P^*)$ (choke = the price where $q = 0$)

PS = $\frac{1}{2} \times Q^* \times (P^* - \text{supply intercept price})$

WORKED EXAMPLE (SAME NUMBERS)

Invert: demand $p = 40 - q/5$ (choke 40); supply $p = 2.5 + q/20$ (intercept 2.5). At $q^*=150$, $p^*=10$:

CS = $\frac{1}{2} \cdot 150 \cdot (40 - 10) = 2250$. **PS** = $\frac{1}{2} \cdot 150 \cdot (10 - 2.5) = 562.5$.

CORE SKILL 2

Elasticity & Revenue

DEFINITION

$\epsilon = |\% \Delta Q / \% \Delta P| = |(dQ/dP) \cdot (P/Q)|$ — always take the absolute value for demand.

ϵ	Name	Quantity response	Raise price → revenue...
> 1	Elastic	very responsive	falls
$= 1$	Unit-elastic	—	unchanged (max)
< 1	Inelastic	barely responds	rises

THE #1 ELASTICITY TRAP

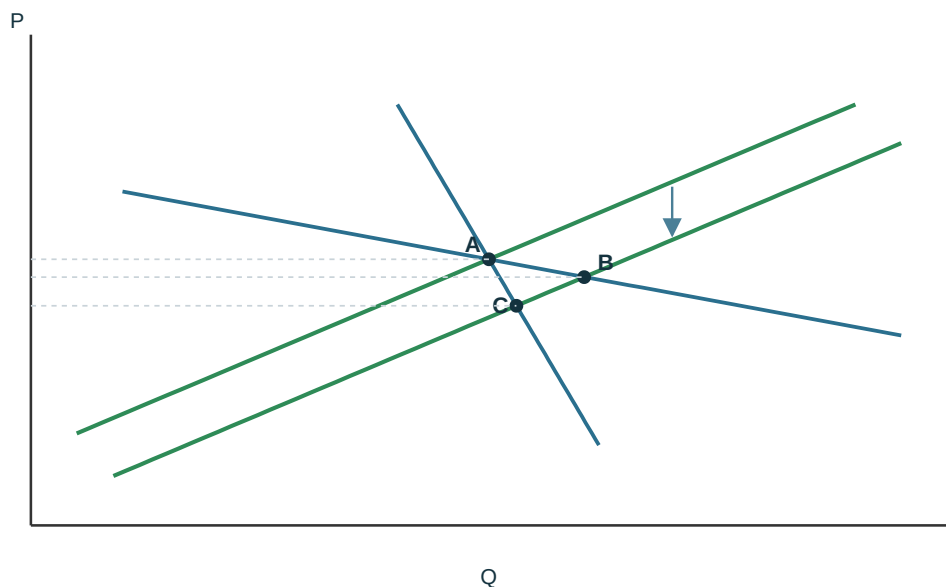
If demand is $q = 200 - 5p$, the **5 is NOT the elasticity** — it's the slope coefficient ($dq/dp = -5$). Elasticity also multiplies by **P/Q**: at $p=10$, $q=150$, $\epsilon = 5 \cdot (10/150) = 1/3$ (inelastic). Always compute P/Q .

WORKED EXAMPLE

Used cars have elasticity 0.6 (inelastic). If price rises 10%, quantity falls only $0.6 \times 10 = 6\%$. Since quantity falls *less* than price rose, **total spending on used cars rises**.

Shifts: elasticity decides price vs quantity

A cost fall (or good harvest) shifts **supply right/down**; a popularity change shifts **demand**. How the new equilibrium splits into a price change vs a quantity change depends on **elasticity**.



Same supply shift, two demands. With **elastic** (flat) demand → price barely falls, quantity jumps (point B). With **inelastic** (steep) demand → price drops a lot, quantity barely moves (point C).

CLASSIC MCQ

"Price will fall by the same amount as costs." **False** — the price change depends on the elasticities of BOTH curves, almost never one-for-one. Also: a price change is a **movement along** a curve, **not** a shift of it.

TWO-SHOCK QUESTIONS (BOTH CURVES MOVE)

If supply shifts left **and** demand shifts left (e.g. fewer shrimp caught + less popular): **quantity falls for certain**, but the **price change is ambiguous** — it depends on the relative size of the two shifts.

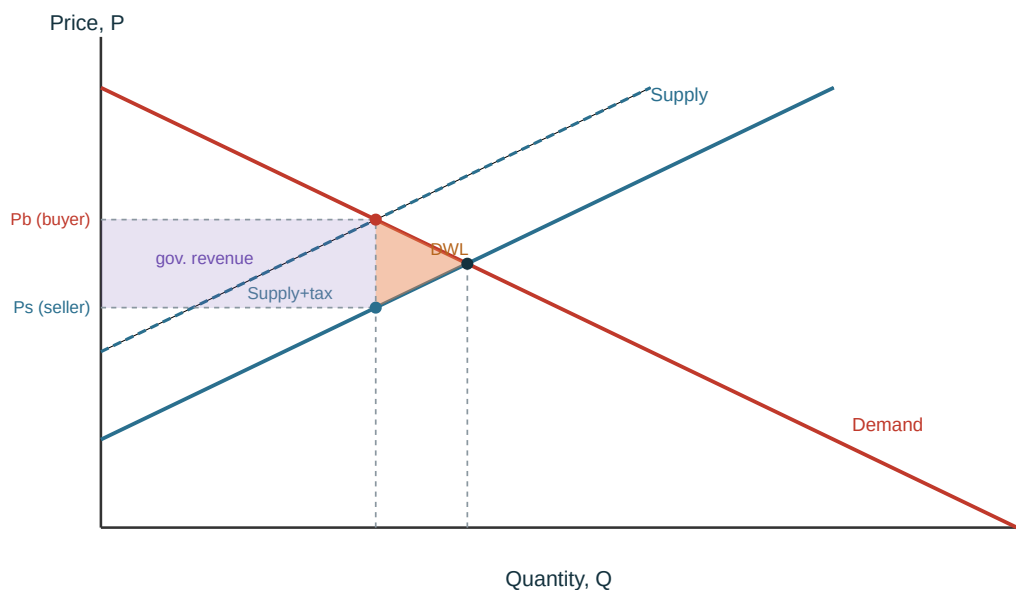
CORE SKILL 3

Taxes — the Wedge

HOW TO SOLVE A SPECIFIC (PER-UNIT) TAX T

Let p = market price. If the **seller** remits the tax: seller keeps $p - t$, buyer pays p . Rewrite supply using $(p - t)$, set new supply = demand, solve.

Buyer price – seller price = t always. Quantity falls.



The tax wedge. Buyer price P_b rises, seller price P_s falls, and $P_b - P_s = t$. Government collects the purple rectangle ($t \times Q$). The orange triangle is the **deadweight loss** — trades that no longer happen.

WORKED EXAMPLE

Supply $q = -80 + 20p$, demand $q = 130 - 10p$. Pre-tax: $p = 7$, $q = 60$. A €3 seller tax $\rightarrow$ supply becomes $q = -80 + 20(p-3) = -140 + 20p$. New: $-140 + 20p = 130 - 10p \rightarrow p$ (buyer) = 9, seller price = 6, $q = 40$. **Buyer price +2, seller price -1.**

INCIDENCE — WHO REALLY PAYS (MEMORISE)

The more INELASTIC side of the market bears the LARGER share of the tax.

- Perfectly inelastic supply $\rightarrow$ sellers bear it all, **consumer price unchanged**.
- Perfectly elastic supply $\rightarrow$ buyers bear it all, market price rises by the full t .
- **It makes no difference whether the buyer or the seller legally pays** — identical prices & quantities. Subsidies = negative taxes.

TRAP

"The impact on the producer price is larger if supply is more elastic." **False** — more elastic supply means sellers bear **less**, so the producer price moves **less**.

UNIT 8

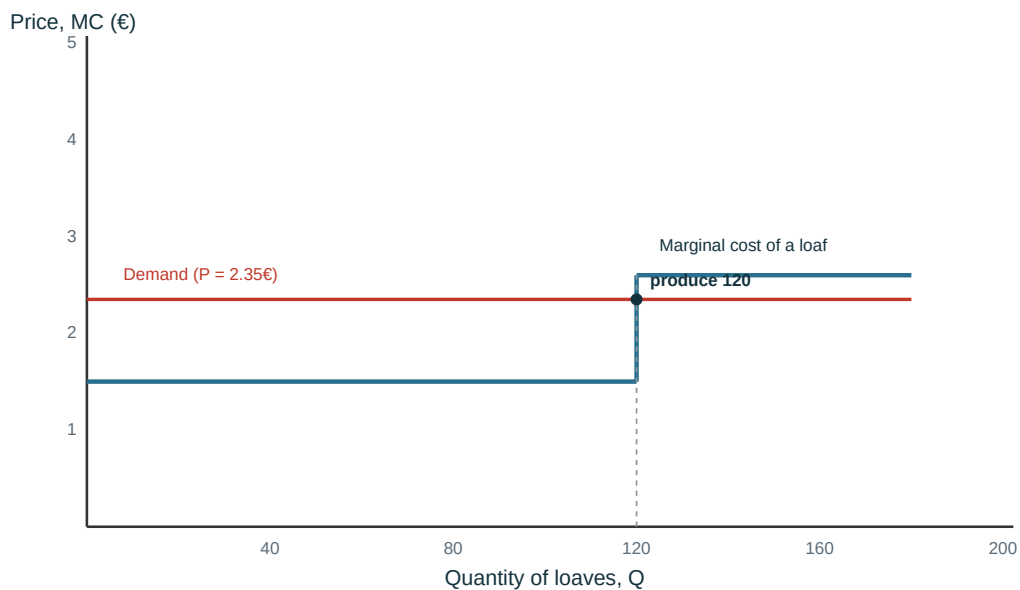
The Competitive (Price-Taking) Firm

PRICE-TAKER

Many firms, identical product → each firm **takes the market price as given** and faces a **horizontal** demand at that price. It only chooses **how much** to produce.

THE RULES (KNOW THESE COLD)

1. Produce where $P = MC$ (because for a price-taker, $MR = P$, and profit-max is $MR = MC$).
2. The firm's **supply curve = its MC curve**, for the part **above AC**.
3. **Shut down / exit** if $P < AC$. Producer surplus = area between price and MC.
4. Market supply = **horizontal sum** of all firms' MC curves.
5. **Long run**: entry drives price down to $P = MC = AC$ → zero economic profit.



Price-taker's choice. Demand is horizontal at the market price. Produce every unit whose price exceeds its marginal cost — here up to where P meets the step in MC.

EFFICIENCY (THE BIG UNIT 8 RESULT)

The competitive equilibrium **maximises total surplus with no deadweight loss** — provided price-taking, complete contracts and **no externalities**. This is what "perfect competition is efficient" means.

TRAPS

· "MR = price" is only true for a **price-taker**. · "Producer surplus is zero in perfect competition" is **false** in general (only if supply is perfectly elastic / constant MC). · Strong economies of scale → markets tend to be **more concentrated**.

UNIT 7

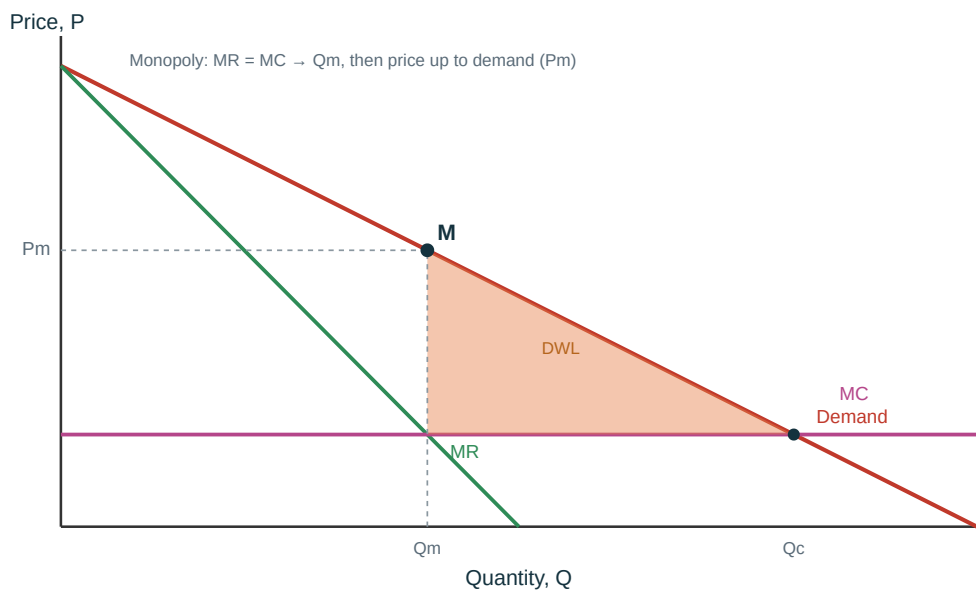
The Firm With Market Power

MARKET POWER

A firm selling a **differentiated** product faces a **downward-sloping** demand curve and can **set a price above marginal cost**. To sell one more unit it must lower the price on **all** units
→ $MR < P$.

THE TWO MUST-KNOW FACTS

Linear demand $P = a - bQ \Rightarrow MR = a - 2bQ$ — same intercept, twice as steep.
Profit max: $MR = MC$. Find Q there, then read the **price up on the demand curve** (so $P > MC$).



Monopoly/market-power picture. MR (twice as steep) meets MC at Q_m ; the price P_m is read up on demand. Because output is restricted below the competitive level Q_c , there is a **deadweight loss** (orange).

WORKED EXAMPLE (MONOPOLY)

Demand $q = 200 - 5p \rightarrow$ inverse $p = 40 - q/5$, so $MR = 40 - 2q/5$. $MC = 2$. Set $MR = MC$:
 $40 - 2q/5 = 2 \rightarrow q = 95$, $p = 40 - 95/5 = 21$. Consumer surplus = $\frac{1}{2} \cdot (40 - 21) \cdot 95 = 902.5$.

THE MARKUP RULE (LERNER)

$(P - MC)/P = 1/\epsilon$ — the price markup equals the **inverse of the demand elasticity**. More elastic demand (bigger ϵ , more competition) → **smaller markup, price closer to MC**. As $\epsilon \rightarrow \infty$ you get perfect competition ($P = MC$, no DWL).

MC & AC (COST SIDE, BOTH UNITS)

$AC = C(Q)/Q$ (slope of the ray from the origin); MC = cost of one more unit (slope of the total-cost curve). If $MC < AC \rightarrow AC$ falling; if $MC > AC \rightarrow AC$ rising. So **MC cuts AC at the minimum of AC**. **Economies of scale** = AC falls as output rises (spreading fixed costs; increasing returns).

Formula Cheat-Sheet & Top Traps

Thing	Formula / rule
Equilibrium	set $q_D = q_S$, solve for p^* , then q^*
Consumer surplus	$\frac{1}{2} \times Q^* \times (\text{demand choke} - P^*)$
Producer surplus	$\frac{1}{2} \times Q^* \times (P^* - \text{supply intercept})$
Elasticity of demand	$\epsilon = (dQ/dP) \cdot (P/Q) $; >1 elastic, <1 inelastic
Revenue & elasticity	inelastic: $P \uparrow \Rightarrow \text{revenue} \uparrow$; elastic: $P \uparrow \Rightarrow \text{revenue} \downarrow$
Specific tax	$P_b - P_s = t$; who pays is irrelevant; DWL created
Tax incidence	less-elastic side bears more of the tax
Competitive firm	$P = MC$; supply = MC above AC ; exit if $P < AC$
Long-run competition	$P = MC = AC$ (zero economic profit)
Market power output	$MR = MC$, then P from demand ; $P > MC$
Linear demand MR	$P = a - bQ \Rightarrow MR = a - 2bQ$ (twice as steep)
Markup (Lerner)	$(P - MC)/P = 1/\epsilon$
AC vs MC	$MC < AC \Rightarrow AC$ falling; they cross at min AC

TOP EXAM TRAPS (FROM PAST PAPERS)

- Slope coefficient $\neq$ elasticity (multiply by P/Q).
- Who legally pays a tax doesn't change the outcome.
- More elastic side of the market bears **less** of a tax.
- A cost cut does **not** cut price one-for-one.
- $MR = P$ only for a price-taker; for market power $MR < P$.
- Setting $P > MC$ (market power) is **not Pareto efficient** (deadweight loss).
- Economies of scale $\neq$ economies of scope (scale = one good, output $\uparrow$; scope = two goods, shared input).
- "Increasing returns / a $>$ proportional output rise" = economies of scale, **not** diseconomies.

TWO-MINUTE SELF-TEST

1. Demand $q = 100 - 2p$, supply $q = -20 + 2p$. Find p^* , q^* , CS, PS. ($p^*=30$, $q^*=40$; $CS=\frac{1}{2}\cdot 40\cdot(50-30)=400$; $PS=\frac{1}{2}\cdot 40\cdot(30-10)=400$.)
2. A monopolist faces $p = 60 - 2q$, $MC = 4$. Find q , p . ($MR = 60 - 4q = 4 \rightarrow q = 14$, $p = 32$.)
3. A €4 tax; demand more inelastic than supply. Who bears more? (*Buyers — the more inelastic side.*)

This is a condensed exam-revision sheet for Units 7 (the firm & market power) and 8 (supply & demand / competitive markets). The full, detailed unit guides follow this section — use them for the diagrams and derivations. Good luck tomorrow!